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**Trivedi et al.**

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(54) **INSERTION APPARATUS FOR USE WITH  
ROTARY MACHINES**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

338,310 A 3/1886 JM  
1,774,986 A 9/1930 MacKenzie  
(Continued)

FOREIGN PATENT DOCUMENTS

CA 2960352 A1 9/2017  
CA 3077622 A1 4/2019  
(Continued)

OTHER PUBLICATIONS

FET20 (Wireless Borescope, Klein Tools, Jan. 2019) (Year: 2019).  
(Continued)

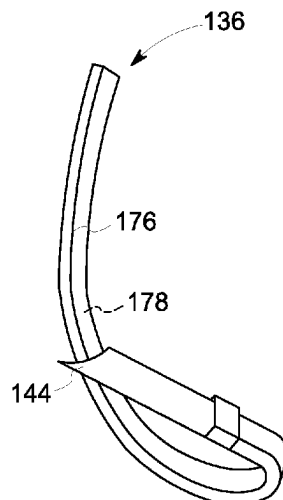
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(57) **ABSTRACT**

An insertion apparatus for use with a rotary machine includes a body extending from an insertion end to a steering end and sized to fit within an annular cavity. The body has a first stiffness and curves along a circumference of the annular cavity as the insertion end travels through the annular cavity. The insertion apparatus also includes a stiffener coupled to the body and extending from the steering end to the insertion end. The stiffener has a second stiffness greater than the first stiffness. The insertion apparatus further includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

**19 Claims, 4 Drawing Sheets**



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|      |                                  |  |           |      |         |            |           |
|------|----------------------------------|--|-----------|------|---------|------------|-----------|
| (51) | <b>Int. Cl.</b>                  |  | 6,123,273 | A    | 9/2000  | Loprinzo   |           |
|      | <b>F01D 5/00</b>                 |  | 6,156,974 | A    | 12/2000 | Blase      |           |
|      | <b>F01D 25/00</b>                |  | 6,213,974 | B1   | 4/2001  | Smith      |           |
|      | <b>F01D 25/28</b>                |  | 6,216,439 | B1   | 4/2001  | Nakamoto   |           |
|      | <b>A61B 17/3207</b>              |  | 6,287,206 | B1   | 9/2001  | Stage      |           |
| (52) | <b>U.S. Cl.</b>                  |  | 6,311,704 | B1   | 11/2001 | Foster     |           |
|      | <b>CPC</b>                       |  | 6,371,148 | B1   | 4/2002  | Tripp      |           |
|      | <b>F01D 25/00</b>                |  | 6,431,824 | B2   | 8/2002  | Schotsch   |           |
|      | <b>(2013.01); F01D 25/285</b>    |  | 6,432,046 | B1 * | 8/2002  | Yarush     | A61B 1/07 |
|      | <b>(2013.01); A61B 17/320758</b> |  |           |      |         |            | 600/179   |
| (56) | <b>References Cited</b>          |  | 6,478,033 | B1   | 11/2002 | Foster     |           |
|      | <b>U.S. PATENT DOCUMENTS</b>     |  | 6,481,195 | B1   | 11/2002 | Blase      |           |
|      |                                  |  | 6,542,230 | B1   | 4/2003  | Luke       |           |
|      |                                  |  | 6,643,877 | B1   | 11/2003 | Amtenbrink |           |
|      |                                  |  | 6,698,456 | B2   | 3/2004  | Neubauer   |           |
|      |                                  |  | 6,783,491 | B2   | 8/2004  | Saadat     |           |
|      |                                  |  | 6,800,016 | B2 * | 10/2004 | Wittenberg | A63H 3/04 |
|      |                                  |  |           |      |         |            | 446/376   |
|      |                                  |  | 6,837,846 | B2   | 1/2005  | Jaffe      |           |
|      |                                  |  | 6,941,974 | B2   | 9/2005  | Utaki      |           |
|      |                                  |  | 6,943,570 | B2   | 9/2005  | Duffy      |           |
|      |                                  |  | 6,955,023 | B2   | 10/2005 | Rotheroe   |           |
|      |                                  |  | 6,957,781 | B2   | 10/2005 | Gowens     |           |
|      |                                  |  | 6,974,411 | B2   | 12/2005 | Belson     |           |
|      |                                  |  | 7,150,416 | B2   | 12/2006 | Martin     |           |
|      |                                  |  | 7,171,279 | B2   | 1/2007  | Buckingham |           |
|      |                                  |  | 7,182,024 | B2   | 2/2007  | Pfeiffer   |           |
|      |                                  |  | 7,182,025 | B2   | 2/2007  | Ghorbel    |           |
|      |                                  |  | 7,185,407 | B2   | 3/2007  | Boyl-Davis |           |
|      |                                  |  | 7,258,521 | B2   | 8/2007  | Guerra     |           |
|      |                                  |  | 7,509,735 | B2   | 3/2009  | Philip     |           |
|      |                                  |  | 7,543,518 | B2   | 6/2009  | Buckingham |           |
|      |                                  |  | 7,559,340 | B2   | 7/2009  | Ikeda      |           |
|      |                                  |  | 7,571,735 | B2   | 8/2009  | Wagner     |           |
|      |                                  |  | 7,654,143 | B2   | 2/2010  | Roney      |           |
|      |                                  |  | 7,662,091 | B2   | 2/2010  | Bagley     |           |
|      |                                  |  | 7,677,181 | B2   | 3/2010  | Boyl-Davis |           |
|      |                                  |  | 7,703,272 | B2   | 4/2010  | Wagner     |           |
|      |                                  |  | 7,707,704 | B2   | 5/2010  | Crocker    |           |
|      |                                  |  | 7,712,301 | B1   | 5/2010  | Wagner     |           |
|      |                                  |  | 7,718,894 | B2   | 5/2010  | Blase      |           |
|      |                                  |  | 7,741,563 | B2   | 6/2010  | Harada     |           |
|      |                                  |  | 7,849,878 | B2   | 12/2010 | Kohler     |           |
|      |                                  |  | 7,854,109 | B2   | 12/2010 | Zubiate    |           |
|      |                                  |  | 7,883,674 | B2   | 2/2011  | Huang      |           |
|      |                                  |  | 8,069,747 | B2   | 12/2011 | Buckingham |           |
|      |                                  |  | 8,096,030 | B2   | 1/2012  | Graichen   |           |
|      |                                  |  | 8,100,031 | B2   | 1/2012  | Zubiate    |           |
|      |                                  |  | 8,125,755 | B2   | 2/2012  | Garcia     |           |
|      |                                  |  | 8,152,934 | B2   | 4/2012  | Lee        |           |
|      |                                  |  | 8,190,294 | B2   | 5/2012  | Sjostrand  |           |
|      |                                  |  | 8,205,522 | B2   | 6/2012  | Buckingham |           |
|      |                                  |  | 8,206,488 | B2   | 6/2012  | Mantkowski |           |
|      |                                  |  | 8,299,785 | B2   | 10/2012 | Bousquet   |           |
|      |                                  |  | 8,303,243 | B2   | 11/2012 | Fish       |           |
|      |                                  |  | 8,327,518 | B2   | 12/2012 | Koerner    |           |
|      |                                  |  | 8,374,722 | B2   | 2/2013  | Buckingham |           |
|      |                                  |  | 8,377,232 | B2   | 2/2013  | Myers      |           |
|      |                                  |  | 8,395,300 | B2   | 3/2013  | Aabloo     |           |
|      |                                  |  | 8,400,501 | B2   | 3/2013  | Heyworth   |           |
|      |                                  |  | 8,409,248 | B2   | 4/2013  | Ginn       |           |
|      |                                  |  | 8,453,533 | B2   | 6/2013  | Ryland     |           |
|      |                                  |  | 8,505,204 | B2   | 8/2013  | Reverchon  |           |
|      |                                  |  | 8,571,711 | B2   | 10/2013 | Jacobsen   |           |
|      |                                  |  | 8,635,849 | B2   | 1/2014  | Tassone    |           |
|      |                                  |  | 8,640,531 | B2   | 2/2014  | Remillard  |           |
|      |                                  |  | 8,674,222 | B2   | 3/2014  | Hsieh      |           |
|      |                                  |  | 8,714,038 | B2   | 5/2014  | Moran      |           |
|      |                                  |  | 8,758,232 | B2   | 6/2014  | Graham     |           |
|      |                                  |  | 8,786,848 | B2   | 7/2014  | Hatcher    |           |
|      |                                  |  | 8,833,826 | B2   | 9/2014  | Garcia     |           |
|      |                                  |  | 8,893,749 | B2   | 11/2014 | Perry      |           |
|      |                                  |  | 8,920,579 | B2   | 12/2014 | Liedtke    |           |
|      |                                  |  | 8,945,096 | B2   | 2/2015  | Zubiate    |           |
|      |                                  |  | 8,959,902 | B2   | 2/2015  | Olivier    |           |
|      |                                  |  | 8,991,163 | B2   | 3/2015  | Olivier    |           |
|      |                                  |  | 8,992,421 | B2   | 3/2015  | Stand      |           |
|      |                                  |  | 8,998,567 | B2   | 4/2015  | Scipio     |           |
|      |                                  |  | 9,016,159 | B2   | 4/2015  | Kell       |           |

# US 12,405,187 B2

Page 3

|      |                         |     |         |              |              |         |           |               |
|------|-------------------------|-----|---------|--------------|--------------|---------|-----------|---------------|
| (56) | <b>References Cited</b> |     |         | 2005/0204489 | A1           | 9/2005  | Velez     |               |
|      | U.S. PATENT DOCUMENTS   |     |         | 2005/0273085 | A1           | 12/2005 | Hinman    |               |
|      |                         |     |         | 2006/0073348 | A1           | 4/2006  | Farmer    |               |
|      |                         |     |         | 2006/0074283 | A1           | 4/2006  | Henderson |               |
|      | 9,016,293               | B2  | 4/2015  | Battaglioli  |              | 4/2006  | Boulais   |               |
|      | 9,028,618               | B2  | 5/2015  | Battaglioli  |              | 6/2006  | Tadano    |               |
|      | 9,127,234               | B2  | 9/2015  | Hughes       |              | 7/2006  | Jacobsen  |               |
|      | 9,149,929               | B2  | 10/2015 | Motzer       |              | 8/2006  | Anhalt    |               |
|      | 9,187,700               | B2  | 11/2015 | Huang        |              | 11/2006 | Moeller   | B24B 19/14    |
|      | 9,220,398               | B2  | 12/2015 | Woodley      |              |         |           | 451/6         |
|      | 9,263,866               | B2  | 2/2016  | Shimizu      | 2008/0066821 | A1      | 3/2008    | Komiya        |
|      | 9,272,425               | B2  | 3/2016  | Garcia       | 2008/0149141 | A1      | 6/2008    | Sales         |
|      | 9,294,737               | B2  | 3/2016  | Hatcher, Jr. | 2008/0161971 | A1      | 7/2008    | Buckingham    |
|      | 9,300,926               | B2  | 3/2016  | Kell         | 2008/0199304 | A1      | 8/2008    | Moran         |
|      | 9,329,377               | B2  | 5/2016  | Kell         | 2008/0218728 | A1      | 9/2008    | Kirschner     |
|      | 9,389,150               | B2  | 7/2016  | Kimpel, Jr.  | 2008/0250769 | A1      | 10/2008   | Wagner        |
|      | 9,399,299               | B2  | 7/2016  | Hermey       | 2009/0084108 | A1      | 4/2009    | Prociw        |
|      | 9,403,244               | B2  | 8/2016  | Rautenberg   | 2009/0084408 | A1      | 4/2009    | Thiemann      |
|      | 9,409,292               | B2  | 8/2016  | Smith        | 2009/0084411 | A1      | 4/2009    | Woodcock      |
|      | 9,435,750               | B2  | 9/2016  | Matsumoto    | 2009/0086014 | A1      | 4/2009    | Lea           |
|      | 9,458,735               | B1  | 10/2016 | Diwinsky     | 2009/0132085 | A1      | 5/2009    | Sjostrand     |
|      | 9,492,906               | B2  | 11/2016 | Rösing       | 2009/0216245 | A1      | 8/2009    | Viola         |
|      | 9,505,125               | B2  | 11/2016 | Zubiate      | 2009/0216374 | A1      | 8/2009    | Low           |
|      | 9,549,663               | B2  | 1/2017  | Larkin       | 2009/0255102 | A1      | 10/2009   | McMasters     |
|      | 9,581,440               | B2  | 2/2017  | Ruhge        | 2009/0255116 | A1      | 10/2009   | McMasters     |
|      | 9,726,628               | B2  | 8/2017  | Ruhge        | 2009/0256007 | A1      | 10/2009   | McMasters     |
|      | 9,733,195               | B2  | 8/2017  | Colletti     | 2009/0320891 | A1      | 12/2009   | Liedtke       |
|      | 9,778,141               | B2  | 10/2017 | Bancalari    | 2010/0030377 | A1      | 2/2010    | Unsworth      |
|      | 9,788,141               | B2  | 10/2017 | Bancalari    | 2010/0037924 | A1      | 2/2010    | Gebhardt      |
|      | 9,857,002               | B2  | 1/2018  | Ott          | 2010/0108107 | A1      | 5/2010    | Mantkowski    |
|      | 9,902,024               | B2  | 2/2018  | Ernst        | 2010/0108377 | A1*     | 5/2010    | Terada        |
|      | 9,909,694               | B2  | 3/2018  | Graham       |              |         |           | B60R 16/0215  |
|      | 9,951,647               | B2  | 4/2018  | Rawson       | 2010/0116292 | A1      | 5/2010    | Wagner        |
|      | 10,060,569              | B2  | 8/2018  | Sivacoe      | 2010/0147330 | A1      | 6/2010    | Kohler        |
|      | 10,085,624              | B2  | 10/2018 | Isoda        | 2010/0160736 | A1      | 6/2010    | Padget        |
|      | 10,197,473              | B2  | 2/2019  | Diwinsky     | 2010/0234988 | A1      | 9/2010    | Buckingham    |
|      | 10,213,919              | B2  | 2/2019  | Axinte       | 2010/0256447 | A1*     | 10/2010   | Dubi          |
|      | 10,238,457              | B2  | 3/2019  | Herrell      |              |         |           | A61B 1/00135  |
|      | 10,265,810              | B2  | 4/2019  | Diwinsky     | 2010/0275404 | A1      | 11/2010   | Myers         |
|      | 10,428,993              | B2  | 10/2019 | Whitefield   | 2011/0030381 | A1      | 2/2011    | Sordyl        |
|      | 10,470,831              | B2  | 11/2019 | Cohen        | 2011/0112527 | A1*     | 5/2011    | Hamilton, Jr. |
|      | 10,488,349              | B2  | 11/2019 | Sibbach      |              |         |           | A61B 17/3468  |
|      | 10,618,162              | B2  | 4/2020  | Norton       | 2011/0152879 | A1      | 6/2011    | Williams      |
|      | 10,639,805              | B2  | 5/2020  | Saraliev     | 2011/0174108 | A1      | 7/2011    | Graham        |
|      | 10,775,315              | B2  | 9/2020  | Mekala       | 2011/0295427 | A1      | 12/2011   | Motzer        |
|      | 10,884,232              | B1  | 1/2021  | Trivedi      | 2011/0303053 | A1      | 12/2011   | Schneider     |
|      | 10,926,403              | B1  | 2/2021  | Asokan       | 2011/0313243 | A1      | 12/2011   | Zubiate       |
|      | 10,962,345              | B2  | 3/2021  | Graham       | 2012/0067158 | A1      | 3/2012    | Kell          |
|      | 10,967,504              | B2  | 4/2021  | Simaan       | 2012/0125164 | A1      | 5/2012    | Kozak         |
|      | 11,371,437              | B2  | 6/2022  | Hawke        | 2012/0167547 | A1      | 7/2012    | Zhang         |
|      | 11,413,763              | B2  | 8/2022  | Lee          | 2012/0184817 | A1      | 7/2012    | Sugiyama      |
|      | 11,419,692              | B2  | 8/2022  | Kim          | 2012/0197241 | A1      | 8/2012    | Golden        |
|      | 11,518,048              | B2  | 12/2022 | Saraliev     | 2012/0260497 | A1      | 10/2012   | White         |
|      | 11,613,003              | B2  | 3/2023  | Graham       | 2012/0279323 | A1      | 11/2012   | Broda         |
|      | 11,692,650              | B2  | 7/2023  | Graham       | 2012/0312103 | A1*     | 12/2012   | Hannott       |
|      | 11,707,819              | B2  | 7/2023  | Graham       |              |         |           | G02B 23/2476  |
|      | 11,752,622              | B2  | 9/2023  | Graham       |              |         |           | 73/865.8      |
|      | 11,787,069              | B2  | 10/2023 | Curle        | 2013/0074879 | A1      | 3/2013    | Battaglioli   |
|      | 11,793,536              | B2  | 10/2023 | Walen        | 2013/0125753 | A1      | 5/2013    | Ono           |
|      | 12,091,981              | B2  | 9/2024  | Foxall       | 2013/0192353 | A1*     | 8/2013    | Hatcher       |
|      | 12,194,620              | B2  | 1/2025  | Graham       |              |         |           | G02B 23/2484  |
|      | 2003/0089267            | A1  | 5/2003  | Ghorbel      |              |         |           | 73/112.01     |
|      | 2003/0171736            | A1* | 9/2003  | Bon          | 2013/0199040 | A1      | 8/2013    | Dudeck        |
|      |                         |     |         |              | 2013/0226033 | A1      | 8/2013    | Eskuri        |
|      |                         |     |         |              | 2013/0255410 | A1      | 10/2013   | Lee           |
|      |                         |     |         |              | 2013/0335530 | A1      | 12/2013   | Hatcher, Jr.  |
|      | 2003/0229420            | A1  | 12/2003 | Buckingham   | 2013/0340559 | A1      | 12/2013   | Danitz        |
|      | 2004/0059191            | A1* | 3/2004  | Krupa        | 2014/0005683 | A1      | 1/2014    | Stand         |
|      |                         |     |         |              | 2014/0012288 | A1      | 1/2014    | Darisse       |
|      | 2004/0138525            | A1  | 7/2004  | Saadat       | 2014/0055596 | A1      | 2/2014    | Hatcher, Jr.  |
|      | 2004/0186350            | A1  | 9/2004  | Brenneman    | 2014/0058364 | A1      | 2/2014    | Donhowe       |
|      | 2004/0193016            | A1* | 9/2004  | Root         | 2014/0069460 | A1      | 3/2014    | Kell          |
|      |                         |     |         |              | 2014/0125791 | A1      | 5/2014    | Arellano      |
|      |                         |     |         |              | 2014/0133269 | A1*     | 5/2014    | Hansen        |
|      |                         |     |         |              |              |         |           | A61B 1/00078  |
|      |                         |     |         |              |              |         |           | 367/7         |
|      | 2004/0249367            | A1  | 12/2004 | Saadat       | 2014/0260755 | A1      | 9/2014    | Dong          |
|      | 2004/0255422            | A1  | 12/2004 | Reback       | 2014/0371764 | A1      | 12/2014   | Oyola         |
|      | 2005/0075538            | A1  | 4/2005  | Banik        | 2015/0032252 | A1      | 1/2015    | Galluzzo      |
|      | 2005/0107667            | A1  | 5/2005  | Danitz       | 2015/0036150 | A1      | 2/2015    | Kobayashi     |
|      | 2005/0124856            | A1* | 6/2005  | Fujikura     | 2015/0064008 | A1      | 3/2015    | Lewis         |
|      |                         |     |         |              | 2015/0159557 | A1      | 6/2015    | Scipio        |
|      | 2005/0148287            | A1  | 7/2005  | Moeller      |              |         |           |               |
|      | 2005/0203340            | A1  | 9/2005  | Butler       |              |         |           |               |

(56)

## References Cited

## U.S. PATENT DOCUMENTS

2015/0202013 A1 7/2015 Teichtmann  
 2015/0233263 A1 8/2015 Battaglioli  
 2015/0246449 A1 9/2015 Sakai  
 2015/0300920 A1 10/2015 Deascanis  
 2015/0338353 A1 11/2015 Bancalari  
 2015/0341600 A1 11/2015 Hatcher, Jr.  
 2015/0360629 A1 12/2015 Sekino  
 2016/0000629 A1 1/2016 Jackson  
 2016/0008990 A1 1/2016 Franz  
 2016/0031078 A1 2/2016 Kapoor  
 2016/0032761 A1 2/2016 Griffiths  
 2016/0040803 A1 2/2016 Steeger  
 2016/0052129 A1 2/2016 Ekas  
 2016/0114488 A1 4/2016 Mascorro Medina  
 2016/0146036 A1 5/2016 Richter  
 2016/0174816 A1 6/2016 Choset  
 2016/0175057 A1 6/2016 Ibach  
 2016/0182776 A1 6/2016 Huang  
 2016/0186602 A1 6/2016 Saenz  
 2016/0339584 A1 11/2016 Esteban Finck  
 2017/0023154 A1 1/2017 Jaeker  
 2017/0095922 A1 4/2017 Licht  
 2017/0100197 A1 4/2017 Zubiate  
 2017/0120457 A1 5/2017 Saraliev  
 2017/0129110 A1 5/2017 Ohm  
 2017/0157719 A1 6/2017 Diwinsky  
 2017/0167289 A1 6/2017 Diwinsky  
 2017/0167953 A1 6/2017 Diwinsky  
 2017/0175569 A1 6/2017 Rawson  
 2017/0191376 A1 7/2017 Eriksen  
 2017/0219814 A1 8/2017 Letter  
 2017/0219815 A1 8/2017 Letter  
 2017/0239762 A1 8/2017 Roberts  
 2017/0274484 A1 9/2017 Roberts  
 2017/0319048 A1 11/2017 Ikeda  
 2017/0328497 A1 11/2017 Dantin  
 2017/0359530 A1 12/2017 Boudin  
 2017/0361470 A1 12/2017 Otero Del Real  
 2018/0058233 A1 3/2018 Norton  
 2018/0071039 A1\* 3/2018 Barnett ..... A61B 34/70  
 2018/0094538 A1 4/2018 Tibbetts  
 2018/0119568 A1 5/2018 Negoescu  
 2018/0149038 A1 5/2018 Eriksen  
 2018/0156062 A1 6/2018 Dede  
 2018/0156132 A1 6/2018 Dede  
 2018/0214220 A1 8/2018 Kan  
 2018/0231162 A1 8/2018 Zeng  
 2018/0242958 A1 8/2018 Dayton  
 2018/0313225 A1 11/2018 Millhaem  
 2018/0361960 A1 12/2018 Yamamoto  
 2019/0022877 A1 1/2019 Akin  
 2019/0046010 A1 2/2019 Tojo  
 2019/0054638 A1 2/2019 Norton  
 2019/0145498 A1 5/2019 Yoon  
 2019/0190190 A1 6/2019 Bourgeas  
 2019/0246878 A1 8/2019 Bodner  
 2019/0277770 A1 9/2019 Mekala  
 2019/0292938 A1 9/2019 Wang  
 2019/0308319 A1 10/2019 Walters  
 2019/0358813 A1 11/2019 Graham  
 2019/0358833 A1 11/2019 Graham  
 2019/0360794 A1 11/2019 Graham  
 2019/0366536 A1 12/2019 Graham  
 2019/0383158 A1 12/2019 Diwinsky  
 2019/0383161 A1 12/2019 Graham  
 2020/0113412 A1 4/2020 Jensen  
 2020/0114497 A1 4/2020 Graham  
 2020/0114505 A1 4/2020 Kikuchi  
 2020/0114528 A1 4/2020 Graham  
 2020/0182345 A1 6/2020 Gu  
 2020/0205908 A1 7/2020 Julian  
 2020/0224552 A1 7/2020 Millhaem  
 2020/0316789 A1 10/2020 Sohmshtetty  
 2020/0319119 A1\* 10/2020 Peters ..... F01D 21/003  
 2020/0323599 A1 10/2020 Kim

2020/0359879 A1 11/2020 Cahill  
 2020/0405142 A1\* 12/2020 Whitaker ..... A61B 1/317  
 2021/0078165 A1 3/2021 Tang  
 2021/0137354 A1 5/2021 Bob  
 2021/0223142 A1\* 7/2021 Sasaki ..... G01M 15/14  
 2021/0229269 A1 7/2021 Graham  
 2021/0229270 A1 7/2021 Graham  
 2021/0231239 A1 7/2021 Graham  
 2021/0285374 A1 9/2021 Hawke  
 2021/0388737 A1 12/2021 Foxall  
 2022/0221706 A1 7/2022 Trivedi  
 2022/0221707 A1 7/2022 Trivedi  
 2022/0290608 A1 9/2022 Hawke  
 2022/0314430 A1 10/2022 Graham  
 2023/0194234 A1 6/2023 Graham  
 2024/0011413 A1 1/2024 Millhaem  
 2024/0280033 A1 8/2024 Foxall  
 2024/0326266 A1 10/2024 Graham  
 2025/0163824 A1 5/2025 Diwinsky

## FOREIGN PATENT DOCUMENTS

CA 3043720 A1 11/2019  
 CN 86101283 A 8/1986  
 CN 1162516 10/1997  
 CN 1050781 C 3/2000  
 CN 1656312 8/2005  
 CN 1678937 10/2005  
 CN 1903517 1/2007  
 CN 101048101 10/2007  
 CN 101048102 10/2007  
 CN 101528111 A 9/2009  
 CN 101881218 11/2010  
 CN 201769177 3/2011  
 CN 201800016 4/2011  
 CN 102257292 11/2011  
 CN 102292013 A 12/2011  
 CN 102687057 9/2012  
 CN 102711585 10/2012  
 CN 102729240 A 10/2012  
 CN 102871636 1/2013  
 CN 203370761 U 1/2014  
 CN 103895012 A 7/2014  
 CN 104175325 12/2014  
 CN 104582909 A 4/2015  
 CN 103639156 B 7/2015  
 CN 104870141 8/2015  
 CN 105144514 12/2015  
 CN 105377116 3/2016  
 CN 105431106 3/2016  
 CN 105436127 A 3/2016  
 CN 105927820 9/2016  
 CN 106113019 11/2016  
 CN 106163431 11/2016  
 CN 106166746 11/2016  
 CN 106427289 A 2/2017  
 CN 106659438 5/2017  
 CN 106988798 7/2017  
 CN 107205622 9/2017  
 CN 107468339 A 12/2017  
 CN 108356747 8/2018  
 CN 108472025 8/2018  
 CN 207941781 10/2018  
 CN 108890656 11/2018  
 CN 108972527 12/2018  
 CN 109068938 12/2018  
 CN 109476019 3/2019  
 CN 109561935 4/2019  
 CN 109716194 5/2019  
 CN 110001286 7/2019  
 CN 110462169 11/2019  
 CN 110529254 12/2019  
 CN 110757412 2/2020  
 CN 111037602 4/2020  
 CN 111486008 8/2020  
 CN 113146599 7/2021  
 CN 113232042 8/2021  
 DE 3504824 8/1986  
 DE 4102211 8/1991

(56)

**References Cited****FOREIGN PATENT DOCUMENTS**

|    |              |      |         |                 |
|----|--------------|------|---------|-----------------|
| DE | 4102211      | A1 * | 8/1991  |                 |
| DE | 29902753     | U    | 6/1999  |                 |
| DE | 10160922     |      | 5/2003  |                 |
| DE | 102019002892 | A1   | 10/2020 |                 |
| DE | 102020106508 |      | 9/2021  |                 |
| EP | 1216797      |      | 6/2002  |                 |
| EP | 1216797      | A1   | 6/2002  |                 |
| EP | 1489269      | A2   | 12/2004 |                 |
| EP | 1574675      | A2   | 9/2005  |                 |
| EP | 2011619      | A2   | 1/2009  |                 |
| EP | 1914010      | A3   | 9/2010  |                 |
| EP | 2237931      | A1   | 10/2010 |                 |
| EP | 2267508      |      | 12/2010 |                 |
| EP | 22677508     | A2   | 12/2010 |                 |
| EP | 1967295      | B1   | 1/2011  |                 |
| EP | 2320262      |      | 5/2011  |                 |
| EP | 2052792      | A3   | 6/2011  |                 |
| EP | 2353739      | A1   | 8/2011  |                 |
| EP | 2375104      |      | 10/2011 |                 |
| EP | 2286933      | B1   | 11/2011 |                 |
| EP | 2275648      | B1   | 3/2012  |                 |
| EP | 1903188      | B1   | 5/2013  |                 |
| EP | 2597273      | A2   | 5/2013  |                 |
| EP | 2629655      |      | 8/2013  |                 |
| EP | 3061923      | A1   | 8/2016  |                 |
| EP | 3072642      | A2   | 9/2016  |                 |
| EP | 1908928      | B1   | 12/2016 |                 |
| EP | 3153604      | A1   | 4/2017  |                 |
| EP | 3176365      | A1   | 6/2017  |                 |
| EP | 3572632      |      | 11/2019 |                 |
| EP | 3572632      | A1   | 11/2019 |                 |
| EP | 3879075      |      | 9/2021  |                 |
| FR | 2956608      | A1   | 8/2011  |                 |
| FR | 2995996      | A1   | 3/2014  |                 |
| FR | 3082136      | A1   | 12/2019 |                 |
| GB | 779248       |      | 7/1957  |                 |
| GB | 1437405      | A    | 5/1976  |                 |
| GB | 2199842      |      | 7/1988  |                 |
| GB | 2228644      | A    | 8/1990  |                 |
| JP | H10146316    | A    | 6/1998  |                 |
| JP | 2006184832   | A    | 7/2006  |                 |
| JP | 2013510339   | A    | 3/2013  |                 |
| MX | 2010013223   | A1 * | 12/2010 | ..... B6B 35/03 |
| NO | 162227       | B    | 8/1989  |                 |
| TW | 201341090    |      | 10/2013 |                 |
| WO | 9116598      | A1   | 10/1991 |                 |
| WO | 0006336      | A1   | 2/2000  |                 |
| WO | 2009081164   | A1   | 7/2009  |                 |
| WO | 2011092891   |      | 8/2011  |                 |
| WO | 2012042921   | A1   | 4/2012  |                 |
| WO | 2012054829   |      | 4/2012  |                 |
| WO | 2012054829   | A2   | 4/2012  |                 |
| WO | 2016063074   | A2   | 4/2016  |                 |
| WO | 2017037723   |      | 3/2017  |                 |
| WO | 2017221982   | A1   | 12/2017 |                 |
| WO | 2018001967   | A1   | 1/2018  |                 |
| WO | 2019076876   |      | 4/2019  |                 |
| WO | 2019076876   | A1   | 4/2019  |                 |
| WO | 2019097688   |      | 5/2019  |                 |
| WO | 2021040376   |      | 3/2021  |                 |
| WO | 2021040376   | A1   | 3/2021  |                 |

**OTHER PUBLICATIONS**

DE4102211A1 English Translation (Year: 1991).\*

Mascarenas, et al., A compliant mechanism for inspecting extremely confined spaces, Smart Materials and Structures, Oct. 26, 2017, vol. 26, pp. 1-16.

Huang et al., In-Situ Continuous Coke Deposit Removal by Catalytic Steam Gasification for Fuel-Cooled Thermal Management, Journal of Engineering for Gas Turbines and Power, vol. 134, Oct. 2012, 8 Pages.

Mascarenas et al., "A Compliant Mechanism for Inspecting Extremely Confined Spaces" Smart Materials and Structures, vol. No. 26, pp. 1-16, Oct. 26, 2017.

U.S. Appl. No. 16/577,331; Notice of Allowance and Fees Due (PTOL-85) mailed Jul. 25, 2022; (pp. 1-5).

U.S. Appl. No. 16/735,191; Non-Final Rejection mailed Aug. 3, 2022; (pp. 1-11).

U.S. Appl. No. 16/750,665; Non-Final Rejection mailed Jul. 20, 2022; (pp. 1-9).

U.S. Appl. No. 16/751,802; Final Rejection mailed Jul. 28, 2022; (pp. 1-9).

Wickham et al., High Heat Flux Surface Coke Deposition and Removal Assessment, Technical Paper, Air Force Research Laboratory, Edwards AFB, Jan. 2015, California, 21 Pages.

U.S. Appl. No. 17/144,487; Non-Final Rejection mailed Aug. 23, 2022; (pp. 1-6).

U.S. Appl. No. 16/751,802; Supplemental Notice of Allowance mailed Feb. 27, 2023; (pp. 1-4).

U.S. Appl. No. 16/898,629; Final Rejection mailed Feb. 28, 2023; (pp. 1-22).

U.S. Appl. No. 16/696,025, filed Nov. 26, 2019.

U.S. Appl. No. 16/750,665, filed Jan. 23, 2020.

U.S. Appl. No. 16/750,743, filed Jan. 23, 2020.

U.S. Appl. No. 16/751,802, filed Jan. 24, 2020.

U.S. Appl. No. 16/813,829, filed Mar. 10, 2020.

U.S. Appl. No. 16/898,629, filed Jun. 11, 2020.

U.S. Appl. No. 17/144,487, filed Jan. 8, 2021.

U.S. Appl. No. 15/986,958, filed May 23, 2018.

US Final Office Action from U.S. Appl. No. 15/986,958 dated Sep. 9, 2020, 10 pgs.

US Non-Final Office Action from U.S. Appl. No. 15/986,958 dated Apr. 23, 2020, 12 pgs.

US Notice of Allowance and Notice of Allowability, dated Nov. 18, 2020, from U.S. Appl. No. 15/986,958, 9 pgs.

U.S. Appl. No. 16/751,802; Non-Final Rejection mailed Feb. 28, 2022; (pp. 1-12).

U.S. Appl. No. 16/750,743; Non-Final Rejection mailed Apr. 27, 2022; (pp. 1-12).

U.S. Appl. No. 16/577,331; Non-Final Rejection mailed Jan. 19, 2022; (pp. 1-8).

U.S. Appl. No. 16/577,331; Notice of Allowance and Fees Due (PTOL-85) mailed Nov. 7, 2022; (pp. 1-5).

U.S. Appl. No. 16/577,331; Notice of Allowance and Fees Due (PTOL-85) mailed Nov. 7, 2022; (pp. 1-16).

U.S. Appl. No. 16/750,665; Notice of Allowance and Fees Due (PTOL-85) mailed Nov. 17, 2022; (pp. 1-5).

U.S. Appl. No. 16/750,743; Final Rejection mailed Nov. 7, 2022; (pp. 1-13).

U.S. Appl. No. 16/750,743; Final Rejection mailed Nov. 7, 2022; (pp. 1-28).

U.S. Appl. No. 16/751,802; Notice of Allowance and Fees Due (PTOL-85) mailed Oct. 19, 2022; (pp. 1-7).

U.S. Appl. No. 16/898,629; Non-Final Rejection mailed Sep. 13, 2022; (pp. 1-14).

U.S. Appl. No. 17/219,577; Notice of Allowance and Fees Due (PTOL-85) mailed Jan. 4, 2023; (pp. 1-5).

U.S. Appl. No. 17/144,487; Final Rejection mailed Jan. 11, 2023; (pp. 1-5).

U.S. Appl. No. 16/577,268; Non-Final Rejection mailed Jan. 20, 2023; (pp. 1-29).

U.S. Appl. No. 16/750,743; Notice of Allowance and Fees Due (PTOL-85) mailed Jan. 23, 2023; (pp. 1-5).

U.S. Appl. No. 17/824,691; Notice of Allowance and Fees Due (PTOL-85) mailed Jun. 21, 2023; (pp. 1-9).

U.S. Appl. No. 16/577,331; Notice of Allowance and Fees Due (PTOL-85) mailed May 26, 2023; (pp. 1-5).

U.S. Appl. No. 16/735,191; Notice of Allowance and Fees Due (PTOL-85) mailed Mar. 27, 2023; (pp. 1-5).

U.S. Appl. No. 16/750,665; Notice of Allowance and Fees Due (PTOL-85) mailed Mar. 24, 2023; (pp. 1-6).

U.S. Appl. No. 16/750,743; Notice of Allowance and Fees Due (PTOL-85) mailed May 16, 2023; (pp. 1-5).

(56)

**References Cited**

## OTHER PUBLICATIONS

U.S. Appl. No. 16/898,629; Non-Final Rejection mailed Jun. 7, 2023; (pp. 1-14).  
 U.S. Appl. No. 17/144,487; Non-Final Rejection mailed Mar. 22, 2023; (pp. 1-6).  
 U.S. Appl. No. 16/898,629; Final Office Action mailed Oct. 10, 2023; (pp. 1-12).  
 U.S. Appl. No. 17/824,691; Notice of Allowance and Fees Due (PTOL-85) mailed Oct. 10, 2023; (pp. 1-8).  
 U.S. Appl. No. 15/812,004, filed Nov. 14, 2017.  
 U.S. Appl. No. 15/914,469, filed Mar. 7, 2018.  
 U.S. Appl. No. 16/008,475; Final Rejection mailed Mar. 2, 2023; (pp. 1-21).  
 U.S. Appl. No. 16/008,475; Final Rejection mailed Aug. 4, 2022; (pp. 1-8).  
 U.S. Appl. No. 16/008,475; Final Rejection mailed Dec. 15, 2023; (pp. 1-24).  
 U.S. Appl. No. 16/008,475; Non-Final Rejection mailed Jun. 15, 2023; (pp. 1-23).  
 U.S. Appl. No. 16/008,475; Non-Final Rejection mailed Oct. 21, 2022; (pp. 1-18).  
 U.S. Appl. No. 17/144,487; Non-Final Rejection mailed Apr. 18, 2024; (pp. 1-12).  
 U.S. Appl. No. 16/008,475; Non-Final Rejection mailed Apr. 22, 2024; (pp. 1-33).  
 U.S. Appl. No. 16/577,268; Notice of Allowance and Fees Due (PTOL-85) mailed Feb. 26, 2024; (pp. 1-12).  
 U.S. Appl. No. 16/898,629; Notice of Allowance and Fees Due (PTOL-85) mailed Jan. 17, 2024; (pp. 1-7).  
 U.S. Appl. No. 17/144,435; Non-Final Rejection mailed Mar. 11, 2024; (pp. 1-12).  
 U.S. Appl. No. 17/144,487; Final Rejection mailed Feb. 12, 2024; (pp. 1-5).  
 U.S. Appl. No. 17/552,848; Requirement for Restriction/Election mailed Feb. 26, 2024; (pp. 1-8).  
 U.S. Appl. No. 18/328,076; Non-Final Rejection mailed Feb. 27, 2024; (pp. 1-14).  
 Bakhshi, M. et al., Tunnel Segmental Lining Geometry, Tolerance and Measurement, Tunnelling & Trenchless Conference, 2018, 10 pp.

U.S. Appl. No. 16/577,268; Non-Final Rejection mailed Jun. 10, 2024; (pp. 1-28).  
 U.S. Appl. No. 17/552,848; Non-Final Rejection mailed Jun. 21, 2024; (pp. 1-38).  
 U.S. Appl. No. 17/552,848; Final Rejection mailed Oct. 28, 2024; (pp. 1-23).  
 U.S. Appl. No. 16/577,268; Notice of Allowance and Fees Due (PTOL-85) mailed Nov. 20, 2024; (pp. 1-15).  
 U.S. Appl. No. 18/328,076; Notice of Allowance and Fees Due (PTOL-85) mailed Dec. 13, 2024; (pp. 1-8).  
 U.S. Appl. No. 18/328,076; Notice of Allowance and Fees Due (PTOL-85) mailed Dec. 26, 2024; (pp. 1-5).  
 U.S. Appl. No. 16/008,475; Final Rejection mailed Nov. 4, 2024; (pp. 1-50).  
 U.S. Appl. No. 16/898,629; Notice of Allowance and Fees Due (PTOL-85) mailed Aug. 2, 2024; (pp. 1-5).  
 U.S. Appl. No. 18/733,557; Application filed Jun. 4, 2024, entitled "Selectively Flexible Extension Tool".  
 U.S. Appl. No. 16/577,268; Notice of Allowance mailed Sep. 13, 2024; (pp. 1-16).  
 U.S. Appl. No. 17/144,487; Final Rejection mailed Sep. 18, 2024; (pp. 1-14).  
 U.S. Appl. No. 16/577,268; Notice of Allowance and Fees Due (PTOL-85) mailed Sep. 13, 2024; (pp. 1-15).  
 U.S. Appl. No. 18/328,076; Final Rejection mailed Sep. 24, 2024; (pp. 1-12).  
 U.S. Appl. No. 17/144,435; Final Rejection mailed Jan. 24, 2025; (pp. 1-12).  
 U.S. Appl. No. 17/144,487; Final Rejection mailed Jan. 23, 2025; (pp. 1-18).  
 U.S. Appl. No. 18/328,076; Notice of Allowance and Fees Due (PTOL-85) mailed Jan. 14, 2025; (pp. 1-5).  
 U.S. Appl. No. 18/328,076; Notice of Allowance and Fees Due (PTOL-85) mailed Feb. 25, 2025; (pp. 1-5).  
 U.S. Appl. No. 17/144,435; Notice of Allowance mailed Apr. 23, 2025; (pp. 1-7).  
 U.S. Appl. No. 17/144,435; Notice of Allowance and Fees Due (PTOL-85) mailed Apr. 23, 2025; (pp. 1-7).  
 U.S. Appl. No. 17/144,487; Notice of Allowance and Fees Due (PTOL-85) mailed Apr. 30, 2025; (pp. 1-24).

\* cited by examiner

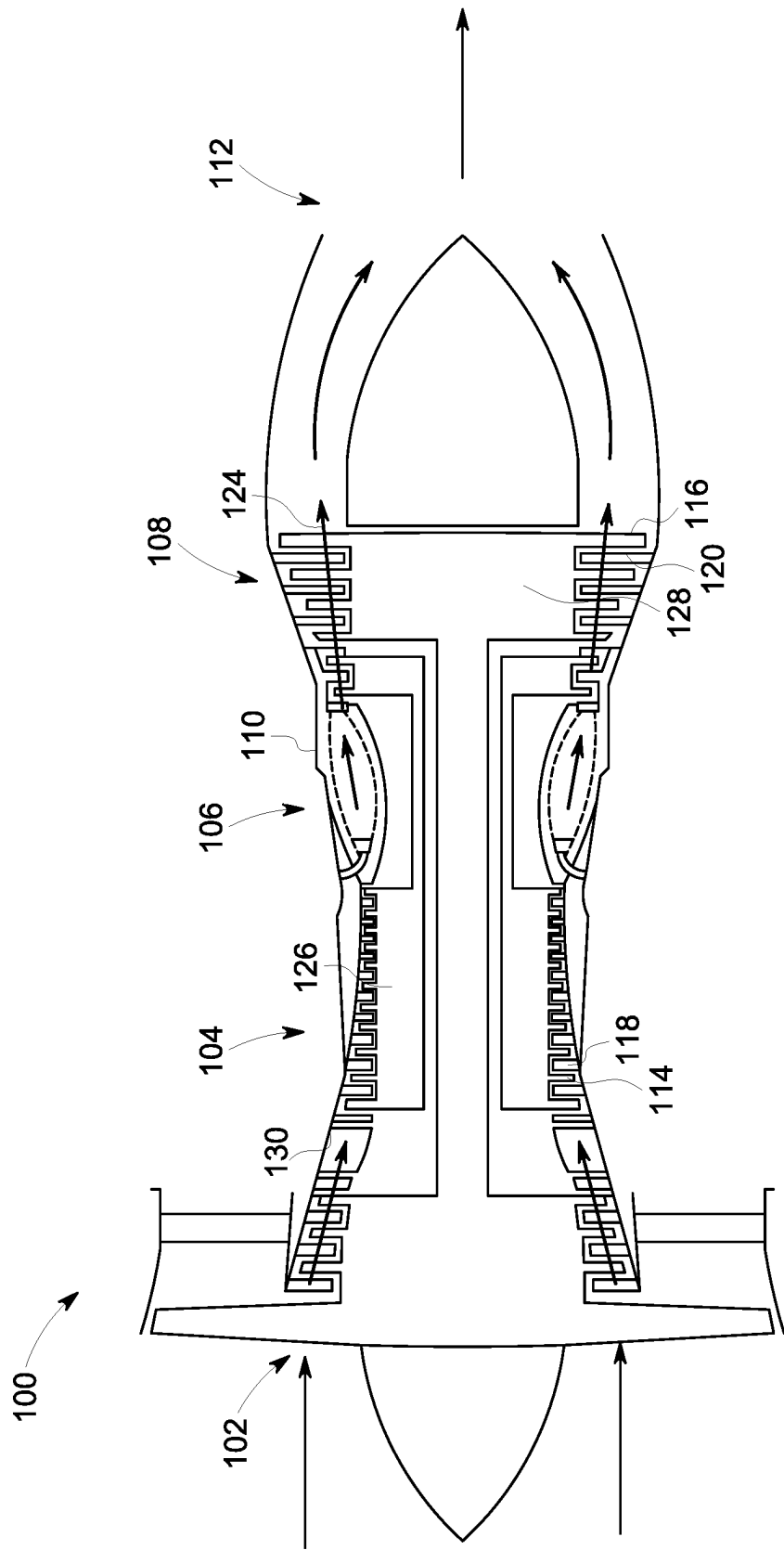


FIG. 1

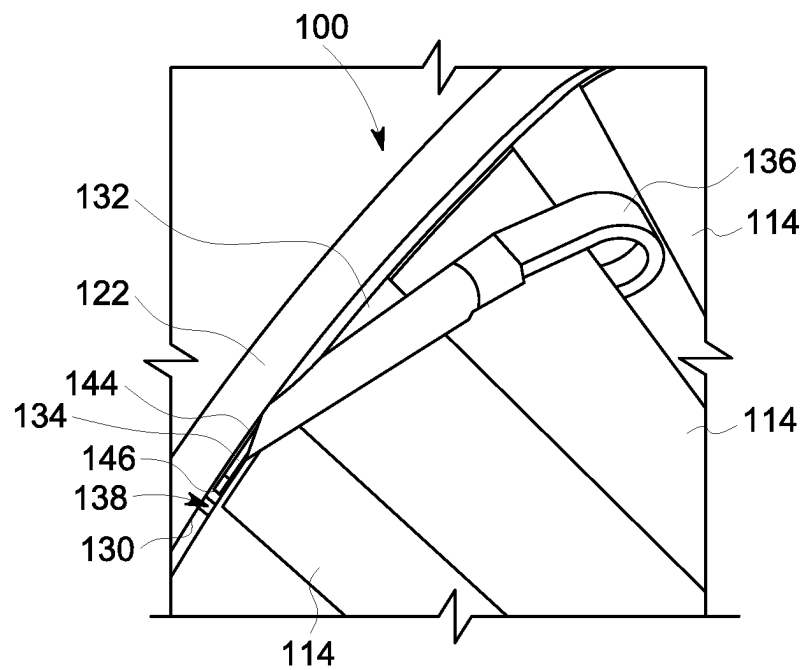


FIG. 2



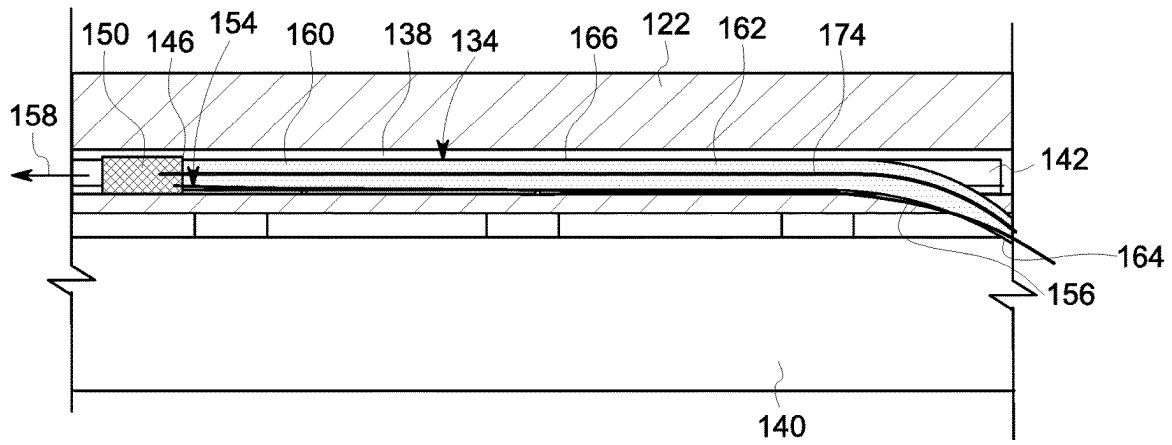


FIG. 3

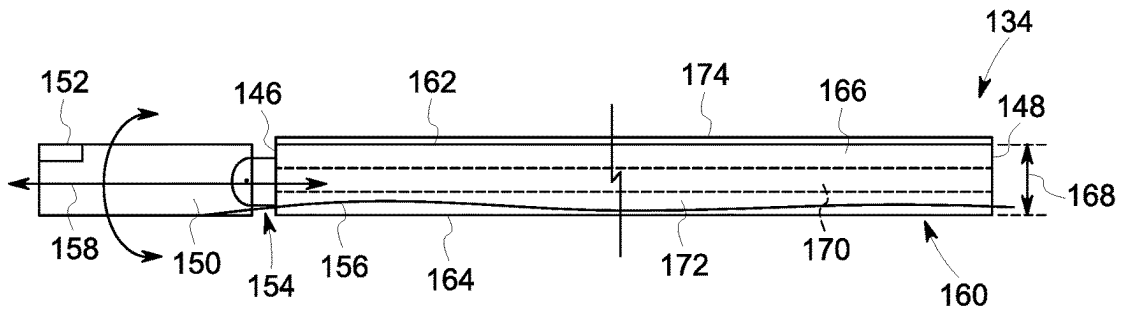


FIG. 4

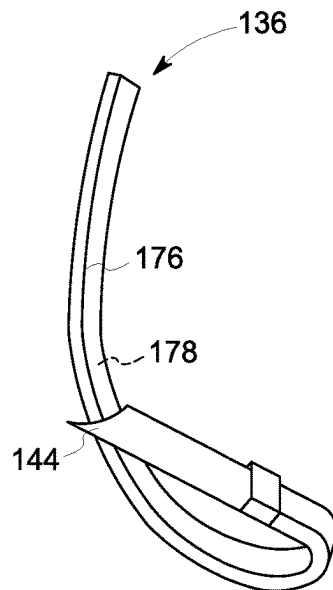


FIG. 5

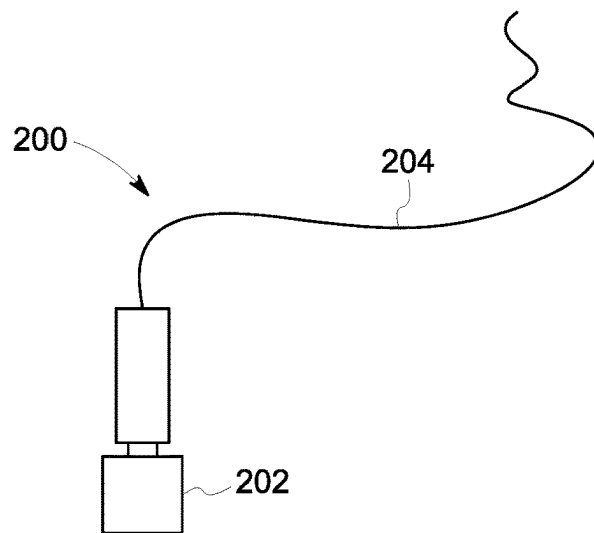


FIG. 6

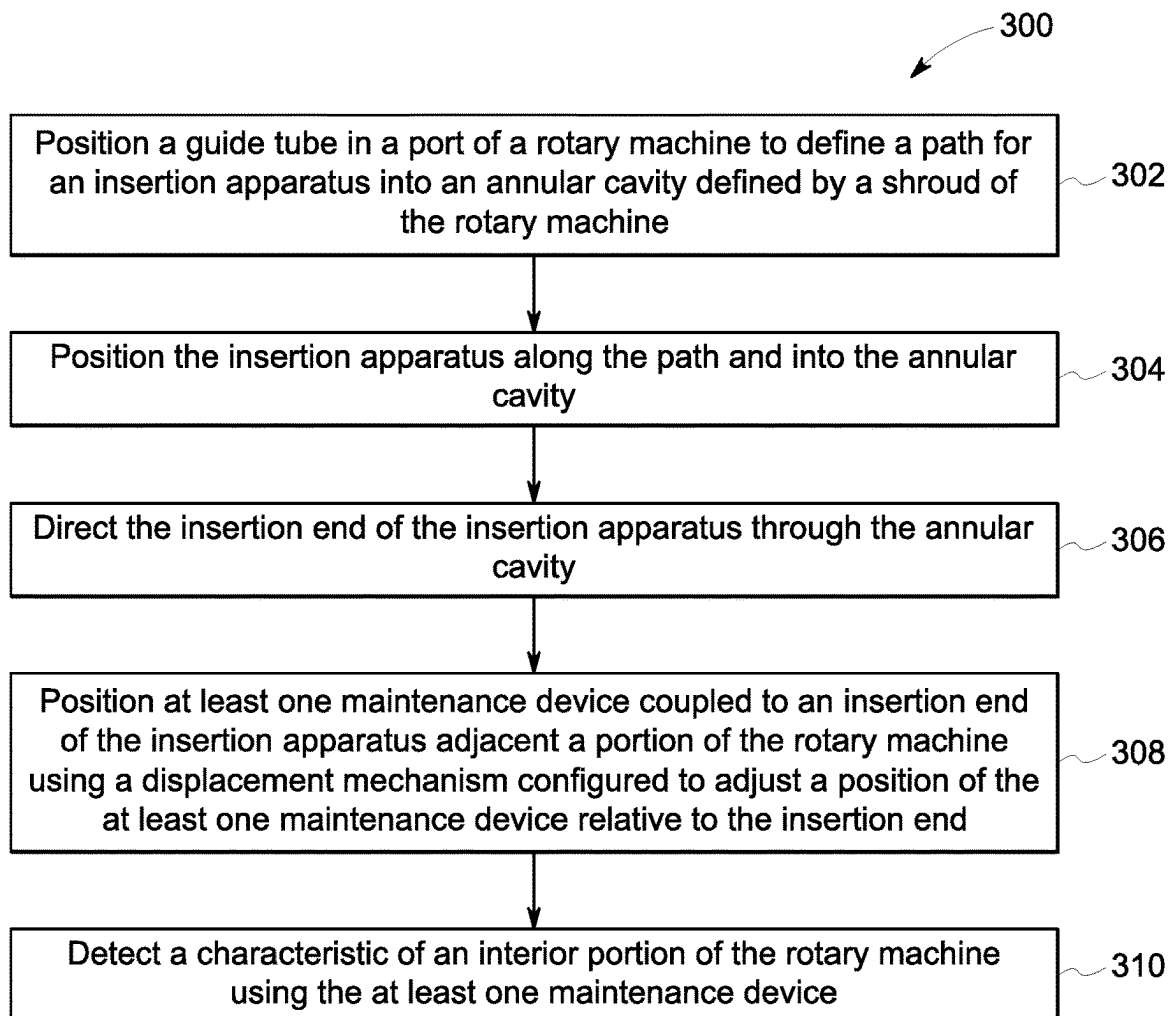


FIG. 7

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## INSERTION APPARATUS FOR USE WITH ROTARY MACHINES

### CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to Indian Patent Application Serial No. 201941040204 filed Oct. 4, 2019, entitled INSERTION APPARATUS FOR USE WITH ROTARY MACHINES, which is hereby incorporated by reference in its entirety.

### BACKGROUND

The field of the disclosure relates generally to an insertion apparatus and, more particularly, to an insertion apparatus for use with rotary machines having annular cavities.

At least some known rotary machines, such as turbines for aircraft engines and gas and steam powered turbines for industrial applications, include an outer case and at least one rotor that carries multiple stages of rotating airfoils, i.e., blades, which rotate with respect to the outer case. In addition, the outer case carries multiple stages of stationary airfoils, i.e., guide vanes. The blades and guide vanes are arranged in alternating stages. In at least some known rotary machines, shrouds are disposed on the radially inner surfaces of a stator to form a ring seal around tips of the blades.

At least some rotary machines are inspected periodically to determine if components of the rotary machines need repair and/or replacement. However, some components of the rotary machine may be difficult to access and inspect without disassembly of the rotary machine. For example, some components, such as clips attaching the shrouds to shroud hangers, are positioned in annular cavities extending circumferentially around the shrouds. However, the size, shape, and location of the annular cavities prevent at least some known inspection apparatus from accessing and inspecting the clips. Moreover, the annular cavities may be obstructed by components, such as the clips, that extend at least partially into the annular cavities.

Accordingly, it is desirable to provide an insertion apparatus for use with a rotary machine that is configured to inspect components positioned along an annular cavity of the rotary machine.

### BRIEF DESCRIPTION

In one aspect, an insertion apparatus for use with a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The insertion apparatus includes an insertion end positionable within the annular cavity and configured to travel through the annular cavity. The insertion apparatus also includes a steering end opposite the insertion end. The insertion apparatus further includes a body extending from the insertion end to the steering end and sized to fit within the annular cavity. The body has a first stiffness and curves along the circumference as the insertion end travels through the annular cavity. The insertion apparatus also includes a stiffener coupled to the body and extending from the steering end to the insertion end. The stiffener has a second stiffness greater than the first stiffness. The insertion apparatus further includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

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In another aspect, a system for use with a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The system includes an insertion apparatus including a body extending from an insertion end to a steering end. The body is sized to fit within the annular cavity and is curved along the circumference. The system also includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body. The system further includes a guide tube positionable in a port of the rotary machine and configured to define a path for the insertion apparatus into the annular cavity. The guide tube is curved to form an at least partially helical path around objects and is sized to extend adjacent an opening into the annular cavity.

In yet another aspect, a method of inspecting a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The method includes positioning a guide tube in a port of the rotary machine to define a path for an insertion apparatus into the annular cavity. The guide tube is curved to form an at least partially helical shape. The method also includes positioning the insertion apparatus along the path and into the annular cavity. The insertion apparatus includes a body having an insertion end and a steering end opposite the insertion end. The body is sized to fit within the annular cavity and is curved along the circumference. The method further includes directing the insertion end of the insertion apparatus through the annular cavity. At least one maintenance device is coupled to the insertion end of the insertion apparatus. The method also includes positioning the at least one maintenance device adjacent a portion of the rotary machine using a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

### DRAWINGS

These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

FIG. 1 is a cross-sectional schematic view of an exemplary rotary machine;

FIG. 2 is a schematic view of a system including an insertion apparatus and a guide tube positioned within a primary cavity of the rotary machine shown in FIG. 1;

FIG. 3 is a schematic view of the insertion apparatus shown in FIG. 2 traveling along the annular cavity of the rotary machine shown in FIG. 1;

FIG. 4 is a side view of a portion of the insertion apparatus shown in FIGS. 2 and 3;

FIG. 5 is a perspective view of the guide tube shown in FIG. 2;

FIG. 6 is a perspective view of an alternative embodiment of a displacement mechanism for use with the insertion apparatus shown in FIGS. 2-4; and

FIG. 7 is a flow chart of an exemplary method of inspecting a rotary machine.

Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of this disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of this disclosure. As such, the drawings are not meant to include all conventional features known by those of

ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

#### DETAILED DESCRIPTION

In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings.

The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

“Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

Embodiments described herein provide a system for use with a rotary machine having an annular cavity. The system includes an insertion apparatus configured to position a maintenance device along the annular cavity of the rotary machine. In addition, in some embodiments, the system includes a guide tube positionable in a port of the rotary machine and configured to guide the insertion apparatus into the annular cavity. For example, the guide tube is curved and has a helical shape. During deployment, the insertion apparatus is inserted along a path defined by the guide tube into the annular cavity and directed along the annular cavity. The insertion apparatus is sized and shaped to fit within and move along the annular cavity. The maintenance device is coupled to an insertion end of the insertion apparatus and is positionable relative to the rotary machine using a displacement mechanism. As a result, the system allows inspection and/or repair of components at locations within the annular cavity of the rotary machine.

FIG. 1 is a cross-sectional schematic view of an exemplary rotary machine. In the exemplary embodiment, the rotary machine includes a turbine assembly 100. In alternative embodiments, the rotary machine includes any assembly. For example, in some embodiments, the rotary machine includes, without limitation, any of the following: a compressor, a blower, a pump, a turbine, a motor, and a generator.

In the exemplary embodiment, turbine assembly 100 includes an inlet 102, a compressor 104, a combustor 106, a turbine 108, an outer case 110, and an exhaust 112. Fluid flows from inlet 102, through compressor 104, through combustor 106, through turbine 108 and is discharged through exhaust 112. Also, in the exemplary embodiment, compressor 104 and turbine 108 include airfoils configured to direct fluid through turbine assembly 100. In particular, compressor 104 and turbine 108 include blades 114, 116 and guide vanes 118, 120. Together, blades 114, 116, guide vanes 118, 120, and shrouds 122 (shown in FIG. 2) define a primary flowpath 124 of turbine assembly 100. This flowpath, combined with a flowpath through combustor 106,

defines a primary cavity within turbine assembly 100. In alternative embodiments, turbine assembly 100 is configured in any manner that enables turbine assembly 100 to operate as described herein.

Blades 114, 116 are operably coupled with rotating shafts 126, 128 such that blades 114, 116 rotate when rotating shafts 126, 128 rotate. Accordingly, blades 114, 116 and rotating shafts 126, 128 form a rotor of turbine assembly 100. Guide vanes 118, 120 and shrouds 122 are stationary components and are coupled to an interior surface 130 of outer case 110. Blades 114, 116 and guide vanes 118, 120 are generally positioned alternately along the rotor axis within turbine assembly 100. In alternative embodiments, compressor 104 and/or turbine 108 includes any airfoils that enable turbine assembly 100 to operate as described herein.

FIG. 2 is a schematic view of a system 132 including an insertion apparatus 134 and a guide tube 136 positioned in the primary cavity of turbine assembly 100. FIG. 3 is a schematic view of insertion apparatus 134 traveling along an annular cavity 138 of turbine assembly 100. Annular cavity 138 extends along a circumference of a respective shroud 122 of turbine assembly 100.

In the exemplary embodiment, shrouds 122 are at least partially supported by a hanger and extend circumferentially around blades 114, 116 and alongside a turbine central frame 140. Clips 142 couple shrouds 122 to hangers and secure shrouds 122 in position during operation of turbine assembly 100. In the exemplary embodiment, clips 142 are C-shaped and extend at least partially into annular cavity 138. Clips 142 are spaced apart along the circumference of each shroud 122. Accordingly, clips 142 are difficult to access from the exterior of turbine assembly 100 or from the primary cavity of turbine assembly 100. Insertion apparatus 134 is sized and shaped to fit within annular cavity 138 and facilitates inspection and/or repair at locations within annular cavity 138 that are difficult to access from an exterior of turbine assembly 100 by conventional means, such as using a borescope tool. Specifically, in contrast to at least some known insertion apparatus, insertion apparatus 134 is configured to extend into annular cavity 138 to facilitate inspection of clips 142.

During operation, insertion apparatus 134 enters turbine assembly 100 through any suitable access port or opening of turbine assembly 100. For example, in some embodiments, insertion apparatus 134 enters and/or exits turbine assembly 100 through any of inlet 102 (shown in FIG. 1), exhaust 112 (shown in FIG. 1), and/or an access port, such as an igniter, borescope, or fuel nozzle port. In the exemplary embodiment, insertion apparatus 134 is sized and shaped to fit within turbine assembly 100 and to travel along annular cavity 138. For example, insertion apparatus 134 has a height, length, and width that are less than a clearance required to fit within annular cavity 138. In alternative embodiments, insertion apparatus 134 is any size and shape that enables insertion apparatus 134 to operate as described herein.

Also, in the exemplary embodiment, guide tube 136 extends through a port of turbine assembly 100 and defines a path for insertion apparatus 134. For example, guide tube 136 defines an interior space sized to receive insertion apparatus 134 and allow insertion apparatus 134 to travel through guide tube 136. Guide tube 136 may be fixed to turbine assembly 100 by a flange (not shown) coupled to a port of turbine assembly 100. For example, in some embodiments, a flange extends around guide tube 136 and is sized to fit onto a port of turbine assembly 100. In alternative

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embodiments, guide tube **136** is coupled to turbine assembly **100** in any manner that enables guide tube **136** to operate as described herein.

In addition, in the exemplary embodiment, guide tube **136** is configured to direct insertion apparatus **134** into annular cavity **138** of turbine assembly **100**. For example, guide tube **136** is curved and defines a curved path for insertion apparatus **134**. Specifically, guide tube **136** is curved in three-dimensions and has a helical shape. Accordingly, guide tube **136** is able to guide insertion apparatus **134** around obstacles in the primary cavity of turbine assembly **100**. In addition, guide tube **136** is sized such that a tip **144** of guide tube **136** is positioned proximate a target area within turbine assembly **100**, e.g., an opening into annular cavity **138**. In alternative embodiments, guide tube **136** is any size and shape that enables guide tube **136** to operate as described herein.

During operation, insertion apparatus **134** is used to inspect and/or repair any interior components of turbine assembly **100**. For example, in some embodiments, insertion apparatus **134** is positioned adjacent a portion of interior surface **130** of turbine assembly **100** within annular cavity **138**. Interior surface **130** may include a surface of clips **142**. In some embodiments, insertion apparatus **134** detects a characteristic of interior surface **130**. For example, in some embodiments, insertion apparatus **134** is used to generate an image of interior surface **130** and the image is examined to determine the condition of turbine assembly **100** and assess whether repairs are necessary. In further embodiments, insertion apparatus **134** includes a sensor that detects characteristics of interior surface **130**. If repairs are necessary, in some embodiments, insertion apparatus **134** is used to repair interior surface **130**. After inspection and/or repair of interior surface **130**, insertion apparatus **134** exits turbine assembly **100** through any suitable access port or opening of turbine assembly **100**, such as via the route of entry.

FIG. 4 is a side view of a portion of insertion apparatus **134**. Insertion apparatus **134** includes an insertion end **146** and a steering end **148** opposite insertion end **146**. Insertion end **146** is positionable within annular cavity **138** (shown in FIG. 3) of turbine assembly **100** (shown in FIG. 1) at target locations, such as adjacent clips **142** (shown in FIG. 3). In some embodiments, insertion end **146** is shaped to facilitate movement of insertion apparatus **134** through annular cavity **138** (shown in FIG. 3) without insertion apparatus **134** being caught on objects. For example, in some embodiments, insertion end **146** is curved. In further embodiments, insertion end **146** is angled, e.g., insertion end **146** has a V-shape.

Also, in the exemplary embodiment, insertion apparatus **134** includes at least one maintenance device **150** coupled to insertion end **146** of insertion apparatus **134** to allow insertion apparatus **134** to perform an inspection and/or repair operation within annular cavity **138** (shown in FIG. 3) of turbine assembly **100** (shown in FIG. 1). In some embodiments, maintenance device **150** includes at least one sensor **152** that is configured to contact surfaces. For example, in some embodiments, sensor **152** is an eddy current sensor. In alternative embodiments, insertion apparatus **134** includes any maintenance device **150** that enables insertion apparatus **134** to operate as described herein. For example, in some embodiments, maintenance device **150** of insertion apparatus **134** includes, without limitation, any of the following: an applicator, a drill, a grinder, a heater, a welding electrode, a sprayer, an optical sensor (e.g., visible, infrared, and/or multi-spectral sensor), a mechanical sensor (e.g., stylus profilometer, coordinate measurement probe, load transducer, linear variable differential transformer), a thermal

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sensor (e.g., pyrometer, thermocouple, resistance temperature detector), a magnetic sensor, an acoustic sensor (e.g., piezoelectric, microphone, ultrasound), and an electromagnetic sensor (e.g., eddy current, potential drop, x-ray).

In addition, in the exemplary embodiment, maintenance device **150** is positionable in a plurality of orientations using a displacement mechanism **154**. For example, in the exemplary embodiment, maintenance device **150** is pivotably coupled to insertion end **146** of insertion apparatus **134**. Displacement mechanism **154** includes a cable **156** extending from maintenance device **150** to steering end **148**. Maintenance device **150** is selectively pivoted by manipulating cable **156**. For example, maintenance device **150** is positionable between a first orientation in which maintenance device **150** is aligned with a translation direction **158** of insertion apparatus **134** and a second orientation in which insertion apparatus **134** extends at an angle relative to translation direction **158**. The ability to position maintenance device **150** in a plurality of orientations facilitates precise positioning of maintenance device **150** relative to a target location. In alternative embodiments, maintenance device **150** is positionable in any manner that enables insertion apparatus **134** to operate as described herein.

Also, in the exemplary embodiment, insertion apparatus **134** includes a body **160** extending from insertion end **146** to steering end **148**. Body **160** is sized and shaped to fit within annular cavity **138** (shown in FIG. 3). For example, body **160** includes a first surface **162**, a second surface **164** opposite first surface **162**, and edges **166** extending along first surface **162** and second surface **164**. First surface **162** and second surface **164** extend from steering end **148** to insertion end **146** and define a thickness **168** of body **160** therebetween. Thickness **168** is less than a width of each of first surface **162** and second surface **164**. For example, in some embodiment, thickness **168** of body **160** is less than about 0.02 inches. In addition, first surface **162** and second surface **164** are substantially smooth, e.g., free of bumps, perforations, folds, or other surface features. Accordingly, body **160** is ribbon-shaped. In alternative embodiments, body **160** has any shape that enables insertion apparatus **134** to function as described herein.

Moreover, in the exemplary embodiment, body **160** is flexible and conforms to a curve of annular cavity **138**. Specifically, body **160** has a first stiffness and is configured to curve in a direction perpendicular to first surface **162** and second surface **164** along the curve of annular cavity **138**. Accordingly, insertion apparatus **134** is able to travel within annular cavity **138** circumferentially along shroud **122** (shown in FIG. 2). Also, body **160** has a length that is greater than a circumference of shroud **122** such that insertion apparatus **134** is configured to extend from the exterior of turbine assembly **100** (shown in FIG. 2), into annular cavity **138** (shown in FIG. 2) and through the entirety of annular cavity **138**. In the exemplary embodiment, body **160** includes at least one conductor **170**, e.g., a flexible metal wire, encased within insulation **172**. In alternative embodiments, insertion apparatus **134** includes any body **160** that enables insertion apparatus **134** to operate as described herein.

In addition, in the exemplary embodiment, insertion apparatus **134** includes a stiffener **174** coupled to and extending along body **160** from steering end **148** to insertion end **146**. In some embodiments, stiffener **174** is coupled to body **160** or formed integrally with body **160**. Stiffener **174** is configured to resist bending of body **160** and facilitate translation and steering of insertion apparatus **134** within annular cavity **138** (shown in FIG. 3). Moreover, stiffener **174**

prevents body 160 from folding as insertion apparatus 134 is moved. Stiffener 174 has a second stiffness greater than the first stiffness of body 160. For example, in some embodiments, stiffener 174 includes a material that has a greater stiffness per unit volume than the material of body 160. In further embodiments, stiffener 174 has a thickness greater than thickness 168 of body 160. In the exemplary embodiment, stiffener 174 includes at least one cable and/or at least one hollow tube extending from steering end 148 to insertion end 146. Stiffener 174 has a width that is less than a width of body 160 and allows some flexing of body 160. In alternative embodiments, insertion apparatus 134 includes any stiffener 174 that enables insertion apparatus 134 to operate as described herein. In further embodiments, stiffener 174 is omitted.

In some embodiments, body 160 and/or stiffener 174 is configured to convey power and communication signals for maintenance device 150. In further embodiments, maintenance device 150 transmits and receives signals and/or power using any wired and/or wireless connections. For example, in some embodiments, a component, such as a harness or tether, extends from maintenance device 150 to the exterior of turbine assembly 100 and provides power to maintenance device 150, allows maintenance device 150 to send and/or receive signals, and/or transmits mechanical force, fluids, or thermal energy to maintenance device 150.

FIG. 5 is a perspective view of guide tube 136. Guide tube 136 includes a body 176 and tip 144. Together body 176 and tip 144 define an interior cavity 178 sized to receive insertion apparatus 134 (shown in FIG. 4). Guide tube 136 is configured to guide insertion apparatus 134 (shown in FIG. 2) toward annular cavity 138 (shown in FIG. 2). Body 176 is substantially rigid relative to insertion apparatus 134 (shown in FIG. 2) such that body 176 does not deform when insertion apparatus 134 travels through guide tube 136. In addition, body 176 is curved in 3-dimensions and forms a helical shape. When positioned in a port of turbine assembly 100 (shown in FIG. 2), guide tube 136 defines a path for insertion apparatus 134 around components of turbine assembly 100, such as blades 114 (shown in FIG. 2), and into annular cavity 138 (shown in FIG. 2). In alternative embodiments, guide tube 136 includes any body 176 that enables guide tube 136 to function as described herein. For example, in some embodiments, body 176 is segmented and the segments of body 176 are positionable relative to each other.

Also, in the exemplary embodiment, tip 144 is pliable and has a stiffness less than the stiffness of body 176. In addition, tip 144 is relatively flexible compared to the rest of guide tube 136. Accordingly, tip 144 may deform when contacted by insertion apparatus 134. Tip 144 facilitates guiding insertion apparatus 134 into the opening of annular cavity 138 when guide tube 136 is positioned proximate annular cavity 138. In alternative embodiments, guide tube 136 includes any tip that enables guide tube 136 to function as described herein.

FIG. 6 is a perspective view of an alternative embodiment of a displacement mechanism 200 for use with the insertion apparatus 134 (shown in FIGS. 2-4). Displacement mechanism 200 is configured to adjust the orientation of maintenance device 150 (shown in FIG. 4) relative to insertion end 146 (shown in FIG. 4) of insertion apparatus 134. For example, displacement mechanism 200 includes an inflatable bladder 202 and a flexible tube 204 configured to deliver a fluid to inflatable bladder 202. Inflatable bladder 202 is configured to couple to maintenance device 150 (shown in FIG. 4). In addition, inflatable bladder 202 is configured to switch between a deflated position and an at

least partially inflated position to adjust the position of maintenance device 150 relative to body 160 (shown in FIG. 4) of insertion apparatus 134 (shown in FIG. 4).

Also, in the exemplary embodiment, flexible tube 204 extends from insertion end 146 (shown in FIG. 4) to steering end 148 (shown in FIG. 4) and is configured to carry fluid such as air, water, and/or any other suitable fluid to inflatable bladder 202. In some embodiments, displacement mechanism 200 includes a valve or other regulatory mechanism to allow control of the fluid flow through flexible tube 204. In some embodiments, flexible tube 204 provides stiffness to body 160 (shown in FIG. 4) and acts as stiffener 174 (shown in FIG. 4) of insertion apparatus 134 (shown in FIG. 4).

FIG. 7 is a flow chart of an exemplary method 300 of inspecting turbine assembly 100 (shown in FIG. 1). In reference to FIGS. 1-3 and 7, method 300 includes positioning 302 guide tube 136 in a port of the rotary machine to define a path for insertion apparatus 134 into annular cavity 138 defined by shroud 122. For example, in some embodiments, a flange fits onto a port of turbine assembly 100 and couples guide tube 136 to turbine assembly 100. Guide tube 136 defines a path from an exterior of turbine assembly 100, through the port, and into annular cavity 138. Guide tube 136 is shaped to curve around obstacles within turbine assembly 100 and is sized such that tip 144 of guide tube 136 is positioned adjacent an opening into annular cavity 138. In alternative embodiments, guide tube 136 defines any path that enables system 132 to operate as described herein.

In addition, method 300 includes positioning 304 insertion apparatus 134 along the path and into annular cavity 138. For example, in some embodiments, insertion apparatus 134 is inserted into the interior cavity 178 of guide tube 136 and moved through guide tube 136. Guide tube 136 guides insertion apparatus 134 around obstacles within turbine assembly 100 and into annular cavity 138 as insertion apparatus 134 is moved along the path defined by guide tube 136. In some embodiments, method 300 includes deforming tip 144 to guide insertion apparatus 134 into annular cavity 138. For example, in some embodiments, tip 144 has a stiffness less than the stiffness of body 176 and allows insertion apparatus 134 to transition smoothly from guide tube 136 and into annular cavity 138.

Also, method 300 includes directing 306 insertion end 146 of insertion apparatus 134 through annular cavity 138. For example, in some embodiments, insertion end 146 of insertion apparatus 134 is directed through annular cavity 138 using a steering interface located at steering end 148. In some embodiments, insertion end 146 is directed by moving body 160 along translation direction 158. Insertion end 146 moves along the circumference of shroud 122 within annular cavity 138 as body 160 is moved in translation direction 158. After insertion end 146 has traveled around substantially the entire circumference of shroud 122 or has passed a desired target location, insertion apparatus 134 may be moved backwards along translation direction 158 within annular cavity 138. In alternative embodiments, insertion apparatus 134 is moved in any manner that enables insertion apparatus 134 to operate as described herein.

Also, method 300 includes positioning 308 maintenance device 150 coupled to insertion end 146 of insertion apparatus 134 adjacent a portion of the rotary machine using displacement mechanism 154. For example, in some embodiments, cable 156 of displacement mechanism 154 is used to pivot maintenance device 150 about insertion end 146 of insertion apparatus 134 such that maintenance device 150 contacts interior surface 130 of clips 142. In further

embodiments, inflatable bladder **202** (shown in FIG. **6**) is inflated/deflated to adjust the position of maintenance device **150**.

Moreover, method **300** includes detecting **310** a characteristic of an interior portion of the rotary machine using maintenance device **150**. For example, in some embodiments, maintenance device **150** includes sensor **152** with electrodes that contact clips **142** within annular cavity **138** to determine a characteristic of clips **142**. Displacement mechanism **154** facilitates maintenance device **150** properly contacting clips **142** or other components of turbine assembly **100**. Accordingly, insertion apparatus **134** is able to provide accurate and reliable information for components along annular cavity **138**.

The above described embodiments provide a system for use with a rotary machine having an annular cavity. The system includes an insertion apparatus configured to position a maintenance device along the annular cavity of the rotary machine. In addition, in some embodiments, the system includes a guide tube positionable in a port of the rotary machine and configured to guide the insertion apparatus into the annular cavity. For example, the guide tube is curved and has a helical shape. During deployment, the insertion apparatus is inserted along a path defined by the guide tube into the annular cavity and directed along the annular cavity. The insertion apparatus is sized and shaped to fit within and move along the annular cavity. The maintenance device is coupled to an insertion end of the insertion apparatus and is positionable relative to the rotary machine using a displacement mechanism. As a result, the system allows inspection and/or repair of components at locations within the annular cavity of the rotary machine.

An exemplary technical effect of the methods, systems, and apparatus described herein includes at least one of: (a) reducing the time to inspect and/or repair a rotary device or other applicable circular mechanism; (b) increasing the accessibility of difficult-to-reach locations within a turbine assembly for inspection and/or in situ repair; (c) reducing the time that circular mechanisms are out of service for maintenance; (d) increasing the precision and/or reliability of inspection and repair of circular mechanisms; (e) reducing unplanned service outages for a circular mechanisms; (f) enhancing data capture for use in quantifying and/or modeling the service condition of at least some components of the circular mechanism; and (g) providing a system and method for inspecting shroud hanger clips within an annular cavity.

Exemplary embodiments of methods and systems for use with rotary machines are not limited to the specific embodiments described herein, but rather, components of systems and/or steps of the methods may be utilized independently and separately from other components and/or steps described herein. For example, the methods and systems may also be used in combination with other systems requiring inspection and/or repair of components, and are not limited to practice with only the systems and methods as described herein. Rather, the exemplary embodiment can be implemented and utilized in connection with many other applications, equipment, and systems that may benefit from using a service apparatus for inspection and/or repair.

Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

This written description uses examples to disclose the embodiments, including the best mode, and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. An insertion apparatus for use with a rotary machine, the rotary machine defining an annular cavity extending along a circumference, said insertion apparatus comprising:
  - an insertion end positionable within the annular cavity and configured to travel through the annular cavity;
  - a steering end opposite said insertion end;
  - a body extending from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body includes a first material having a first material stiffness and curves along the circumference as said insertion end travels through the annular cavity;
  - a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener includes a second material having a second material stiffness that is greater than the first material stiffness and is configured to resist bending of said body;
  - wherein the stiffener is fixed adjacent to one edge of the body;
  - wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the body that does not correspond to the stiffener during insertion into the annular cavity;
  - wherein the stiffener is configured to resist bending and prevent the body from folding when being translated into the annular cavity;
  - wherein the body and stiffener comprise a uniform and uninterrupted cross-section extending from the insertion end to the steering end;
  - wherein (1) arrangement of the stiffener fixed and adjacent to one edge of the body and (2) integral formation of the body with the stiffener are cooperatively effective to provide the body with an overall stiffness, the overall stiffness being constant and unchanged prior to deformation of the body and after deformation of the body as the body curves along the circumference during insertion and movement of the body through the annular cavity;
  - wherein the stiffener is uncontrolled during insertion and movement of the body through the annular cavity;
  - at least one maintenance device coupled to said insertion end of said insertion apparatus;
  - a displacement mechanism configured to adjust a position of said at least one maintenance device relative to said body;
  - a guide tube having a segmented body including segments that are positionable relative to each other, the body and stiffener axially movable through the segments of the guide tube; and
  - wherein the guide tube is curved to form an at least partially helical shape.
2. The insertion apparatus in accordance with claim 1, wherein said body comprises a first surface and a second surface opposite said first surface, said first surface and said

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second surface extending from said steering end to said insertion end, said body having a thickness defined between said first surface and said second surface, wherein the thickness is less than a width of said first surface and said second surface, and wherein said body is configured to curve 5 in a direction perpendicular to said first surface and said second surface.

3. The insertion apparatus in accordance with claim 1, wherein said at least one maintenance device is pivotably coupled to said insertion end, and wherein said displacement mechanism is configured to pivot said at least one maintenance device about said insertion end. 10

4. The insertion apparatus in accordance with claim 1, wherein said body is configured to convey power and communication signals for said at least one maintenance 15 device.

5. The insertion apparatus in accordance with claim 1, wherein said at least one maintenance device comprises at least one of the following: an optical sensor, a mechanical sensor, a thermal sensor, a magnetic sensor, an acoustic 20 sensor, and an electromagnetic sensor.

6. The insertion apparatus in accordance with claim 1, wherein the stiffener has a stiffener thickness greater than a thickness of the body.

7. The insertion apparatus in accordance with claim 1, 25 wherein the body is a ribbon-shaped body.

8. The insertion apparatus in accordance with claim 7, wherein the body has a thickness less than about 0.02 inches.

9. The insertion apparatus in accordance with claim 1, wherein the guide tube has a substantially rigid body and a 30 pliable tip at a guide tube insertion end.

10. The insertion apparatus in accordance with claim 1, wherein the body and the guide tube both comprise a substantially rectangular cross-section.

11. An insertion apparatus for use with a rotary machine, 35 the rotary machine defining an annular cavity extending along a circumference, said insertion apparatus comprising:

an insertion end positionable within the annular cavity and configured to travel through the annular cavity;

a steering end opposite said insertion end; 40

a body extending from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body has a first stiffness, wherein said body comprises a first continuous surface and a second continuous surface opposite the first continuous surface 45 that curve along the circumference as said insertion end travels through the annular cavity;

a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener has a second 50 stiffness greater than the first stiffness and is configured to resist bending of said body;

wherein the stiffener is fixed adjacent to one edge of the body;

wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the body that does not correspond to the stiffener during insertion into the annular cavity;

wherein the stiffener is configured to resist bending and prevent the body from folding when being translated 60 into the annular cavity;

wherein the body and stiffener comprise a uniform and uninterrupted cross-section extending from the insertion end to the steering end;

wherein (1) arrangement of the stiffener fixed and adjacent to one edge of the body and (2) integral formation 65 of the body with the stiffener are cooperatively effective

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to provide the body with an overall stiffness, the overall stiffness being constant and unchanged prior to deformation of the body and after deformation of the body as the body curves along the circumference during insertion and movement of the body through the annular cavity;

wherein the stiffener is uncontrolled during insertion and movement of the body through the annular cavity;

at least one maintenance device coupled to said insertion end of said insertion apparatus;

a displacement mechanism configured to adjust a position of said at least one maintenance device relative to said body; and

a guide tube having a segmented body including segments that are positionable relative to each other, the body and stiffener axially movable through the segments of the guide tube,

wherein the guide tube is curved to form an at least partially helical shape; and

wherein said first continuous surface and said second continuous surface extend from said steering end to said insertion end, said body having a thickness defined between said first continuous surface and said second continuous surface less than about 0.02 inches, wherein the thickness is less than a width of said first continuous surface and said second continuous surface, the width extending in a direction transverse a longitudinal axis of the body such that the body is a ribbon-shaped body, and wherein said body is configured to curve in a direction perpendicular to said first continuous surface and said second continuous surface, wherein the stiffener has a stiffener thickness greater than the thickness of the body.

12. The insertion apparatus in accordance with claim 11, wherein the body and stiffener are axially movable through the segments of the guide tube.

13. The insertion apparatus in accordance with claim 12, wherein the guide tube has a substantially rigid body and a pliable tip at a guide tube insertion end.

14. A method of inspecting a rotary machine, the rotary machine defining an annular cavity extending along a circumference, said method comprising:

positioning a guide tube in a port of the rotary machine to define a path for an insertion apparatus into the annular cavity, the guide tube having a substantially rigid body and a pliable tip at a guide tube insertion end, the insertion apparatus comprising:

an insertion end positionable within the annular cavity and configured to travel through the annular cavity;

a steering end opposite said insertion end;

a body extending continuously from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body is formed of a flexible material having a first stiffness and curves along the circumference as said insertion end travels through the annular cavity;

a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener has a second stiffness that is greater than the first stiffness and is configured to resist bending of said body, wherein the guide tube is curved to form an at least partially helical shape;

wherein the stiffener is fixed adjacent to one edge of the body;

wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the



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body that does not correspond to the stiffener during  
inertion into the annular cavity;  
wherein the stiffener is configured to resist bending and  
prevent the body from folding when being translated  
into the annular cavity;  
wherein the body and stiffener comprise a uniform and  
uninterrupted cross-section extending from the inser-  
tion end to the steering end;  
at least one maintenance device coupled to said inser-  
tion end of said insertion apparatus; and  
a displacement mechanism configured to adjust a position  
of said at least one maintenance device relative to said  
body;  
positioning the insertion apparatus along the path and into  
the annular cavity;  
guiding the body of the insertion apparatus through the  
guide tube such that the substantially rigid body of the  
guide tube does not deform during guiding of the  
insertion apparatus therethrough and the pliable tip is  
deformable when contacted by the insertion apparatus;  
directing the insertion end of the insertion apparatus  
through the annular cavity without controlling the  
stiffener; and  
positioning the at least one maintenance device adjacent  
a portion of the rotary machine using the displacement  
mechanism;  
wherein (1) arrangement of the stiffener fixed and adja-  
cent to one edge of the body and (2) the integral  
formation of the body with the stiffener are coopera-  
tively effective to provide the body with an overall

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stiffness, the overall stiffness being constant and  
unchanged prior to deformation of the body and after  
deformation of the body as the body curves along the  
circumference during insertion and movement of the  
body through the annular cavity.

15 15. The method in accordance with claim 14, wherein said  
at least one maintenance device comprises at least one of the  
following: a mechanical sensor, a thermal sensor, a magnetic  
sensor, an acoustic sensor, and an electromagnetic sensor.

10 16. The method in accordance with claim 14, wherein said  
guide tube includes a plurality of positionable segments.

17. The method in accordance with claim 14, wherein said  
guide tube is coupled to the port and is sized and shaped to  
extend from the port to the annular cavity, said guide tube  
15 configured to guide said insertion apparatus through the port  
and into the annular cavity.

18. The method in accordance with claim 14, wherein  
positioning the at least one maintenance device adjacent a  
portion of the rotary machine using a displacement mecha-  
nism configured to adjust a position of the at least one  
20 maintenance device relative to the body comprises pivoting  
the at least one maintenance device about the insertion end  
of the insertion apparatus.

19. The method in accordance with claim 14, wherein  
25 positioning the at least one maintenance device adjacent a  
portion of the rotary machine using a displacement mecha-  
nism configured to adjust a position of the at least one  
maintenance device relative to the body comprises inflating  
a bladder coupled to the at least one maintenance device.

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